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AE 429 | CubeSat Timeline & Details

Project *SINAG*: A Multi-Payload Educational CubeSat for Weather Monitoring, Store-and-Forward Communication, and Radiation Dosimetry

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Authors

Project SINAG CubeSat Development Team

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Chapter 1 – Introduction

1.1 Background of the Study

Project SINAG is a 2U CubeSat project developed to educate and demonstrate emerging technologies. This initiative contributes to the growing array of nanosatellite projects both in the Philippines and globally. CubeSats are compact, standardized satellites that enable universities and emerging space-faring nations to access space at a reduced cost, while also providing practical experience in systems engineering, mission design, and operations in the space environment. The Philippine CubeSat missions, such as Maya-1 and Maya-2, have demonstrated the potential of small satellites to enhance local capabilities, facilitate remote sensing, conduct communication experiments, and provide training in space science and engineering.

1.2 Project SINAG Overview

Project SINAG (Student-led Integrated Nanosatellite for Advancement and Guidance) is an aerospace engineering initiative led by students, dedicated to the design, development, and integration of a 2U CubeSat multi-payload system for environmental monitoring, communications testing, and radiation dosimetry. The name SINAG, originating from the Filipino term meaning "ray of light," embodies enlightenment, innovation, and progress; principles that align with both the project's technical goals and its educational purpose. Inspired by the sun beams of the Philippine flag, Project SINAG represents the commitment to national progress through accessible and locally developed space technology.

The CubeSat payload architecture consists of three interconnected missions: 1) Weather monitoring payload that collects data on the atmosphere and the environment; 2) Store-and-forward communication payload that uses LoRa technology to demonstrate efficient low-power data transmission; 3) The LEO Radiation Dosimetry Payload which measures radiation exposure. These payloads are integrated into a single nanosatellite platform to demonstrate efficient subsystem coordination while maintaining compliance with standard CubeSat form factor and systems engineering constraints.

1.3 Mission Objectives

The primary mission objectives of Project SINAG are to:

- Demonstrate in-orbit weather monitoring: Collect and downlink real-time atmospheric data (e.g., temperature, pressure, humidity) using the onboard weather station payload to validate low-cost environmental sensing in low Earth orbit (LEO).
- Validate LoRa-based store-and-forward communications: Test long-range, low-power LoRa transceivers for relaying small data packets from ground assets or other nanosatellites, achieving at least 90% successful packet reception during operational passes.
- Measure radiation environment: Quantify ionizing radiation levels with the dosimetry payload, generating a dataset for analysis of single-event effects on CubeSat electronics and contributing to radiation hardening standards for Philippine nanosatellites.
- Build student expertise and national capacity: Train multidisciplinary teams in full-cycle CubeSat development, from design to operations, while advancing local nanosatellite technology for disaster monitoring and space education

1.4 Scope and Limitations

Project SINAG aims to design, integrate, and undergo ground testing of a 2U CubeSat prototype (about 10 cm × 10 cm × 22.7 cm, mass under 2.5 kg) in just 4 months. Full subsystem integration (EPS, ADCS, OBC, structure, and ground station simulator), payload development (weather station, LoRa communication, and radiation dosimetry), and rigorous ground verification (vibration testing, radio frequency characterization, EMC tests, and functional payload demonstrations in controlled environments) are included. Students design, prototype, assemble, and test CubeSat Design Specification (CDS) Rev. 14 and Philippine Space Agency (PhilSA) standards to attain technology readiness level. The ground-only focus and 4-month timeline exclude space environmental testing (e.g., radiation exposure or microgravity simulation) and launch integration. Reliance on commercial off-the-shelf (COTS) components may limit performance validation under full space-like conditions.

1.5 Significance of the Project

This study supports governmental and academic initiatives that encourage higher education institutions to develop CubeSats as a means of training future space engineers and demonstrating cost-effective space technologies. By utilizing commercially available components and established CubeSat standards, Project SINAG contributes to the advancement of local aerospace engineering expertise and evaluates the potential of small satellites for environmental monitoring and disaster risk reduction.

Additionally, this study will be significant in providing practical instruction in satellite systems design, promoting interdisciplinary collaboration, and strengthening mission planning skills among students. The outcomes of this project are expected to contribute to the growth of regional aerospace capabilities, foster innovation, and support the sustainable and inclusive development of space technology.

1.5 Project Organization and Team Structure

Project SINAG uses a systems-based approach, with teams for Payload, Communication & Ground Station, EPS, ADCS, Structure & Mechanism, and OBC. Project managers coordinate overall schedule and ensure subsystem alignment with mission objectives. Each team handles the design, construction, and testing of its subsystem while ensuring compatibility with the satellite as a whole.

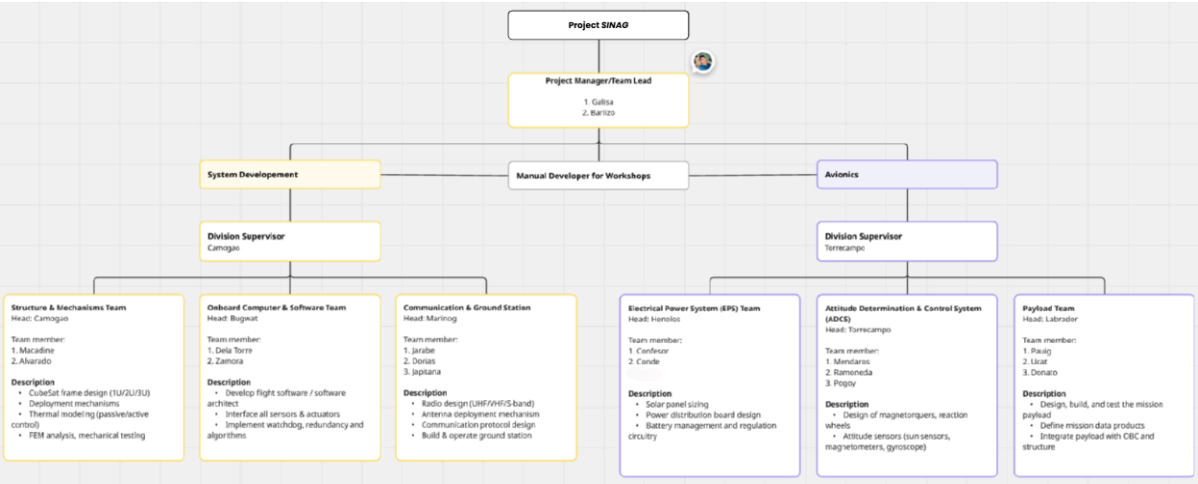

Figure 1. *Organizational structure of Project SINAG and its subsystem teams*

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Chapter 2 – Payload Subsystem

2.1 Payload Team Organization and Roles

This team is composed of four members: Francine May Labrador as the Payload Team Leader, Dan Pauig for Weather Station Payload Engineer, Nathaniel Donato for Store and Forward Payload Engineer, and Hendrix M. Ucat as the Radiation Payload Engineer. Each role is integral to ensuring the success of the Cube Satellite Payload

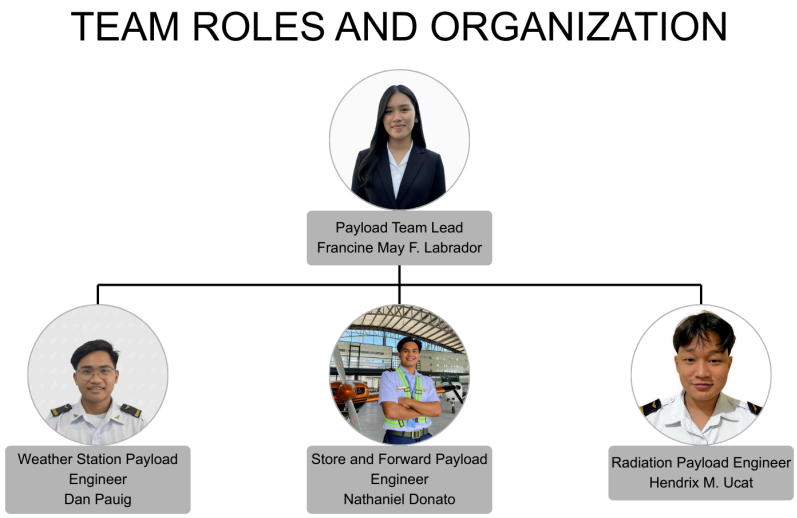


Figure 2. *Team Roles and Organizational Chart*

Table 1. *Payload Team Roles and Responsibilities*

Name	Role	Responsibilities
Francine May Labrador	Payload Team Leader	Oversees the payload mission progress, timeline, team coordination, and overall execution of the payload development.
Dan Pauig	Weather Station Payload Engineer	Leads the sensor selection, calibration plan, and mechanical/electrical interfaces for weather instruments.
Nathaniel Donato	Store and Forward Payload Engineer	Designs the data collection concept from remote ground nodes, defines packet structure, memory needs, and works closely with the Comms team on modulation and link budget.
Hendrix M. Ucat	Radiation Payload Engineer	Leads the radiation sensor selection, shielding trades, and dose-to-data conversion algorithms.

2.2 Payload Mission Overview

The CubeSat payload has three missions that share power, data handling, and structure: a main weather station that sends heat-index SMS alerts via GSM, a secondary LoRa store-and-forward communication experiment that simulates LEO conditions, and a tertiary radiation dosimeter that uses a Geiger counter. A central microcontroller handles sensor data, microSD storage, sending and receiving data, and satellite bus connections.

2.3 Payload Missions

2.3.1 Primary Payload: Weather Station and Heat Index Alert System

The primary mission of the CubeSat is to make a weather station that measures temperature, pressure, lightning, and humidity using the five sensors listed in Table 1. Using the processed weather station data, the system will continuously compute the heat index, and whenever the calculated value exceeds the predefined safety threshold, the program will automatically trigger the SIM800L GSM/GPRS module to send an SMS heat-warning alert to designated mobile numbers. A microSD card module is included to store the measured weather parameters and calculated heat index values for later retrieval and analysis.

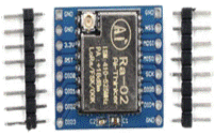
Table 2. *Primary CubeSat Sensors*

Item	Description	Figures	Functions
1.	MS5611 Pressure and Temperature Sensor		Measures air pressure and temperature.
2.	AS3935 Lightning Sensor		Detects nearby lightning activity.
3.	HTU21D Humidity and Temperature Sensor		Measures air humidity and temperature.
4.	SIM800L GSM/GPRS Communication Module		Send SMS messages.
5.	MicroSD Card Module		Stores logged data.

2.3.2 Secondary Payload: LoRa Store-and-Forward Communication Experiment

The secondary mission of the CubeSat is to analyze the link budget and signal integrity of a low-cost LoRa-based store-and-forward communication relay while operating in a high-velocity, high-Doppler environment that simulates Low Earth Orbit conditions, using an SX1278 or SX1262 LoRa transceiver module as the radio "modem" for long-range data transmission and reception, a half-wave dipole antenna made from simple copper wire or tape measure to provide an omnidirectional radiation pattern for maintaining reliable links with ground stations during the simulated satellite pass, a microSD card module for non-volatile storage of received and transmitted messages to enable store-and-forward operation and detailed post-mission analysis of packet integrity and timing, and a payload microcontroller that functions as the command and data handling unit by validating packets, managing memory, assigning packet IDs, and interfacing with any auxiliary science or telemetry sensors that connect to the shared communication bus.

Table 3. *Secondary CubeSat Sensors*

Item	Description	Figures	Functions
1.	SX1278 / SX1262 LoRa Transceiver Module		The radio "modem" that transmits and receives LoRa-modulated signals with high sensitivity for long-range store-and-forward communication.
2.	microSD Card Module		Stores received and outgoing messages in non-volatile memory to enable store-and-forward operation and post-mission data analysis.

2.3.3 Tertiary Payload: Radiation Dosimetry Payload

CubeSat's Low Earth Orbit Tertiary Mission payload measures ionizing radiation. Geiger–Müller tubes on Geiger counter electronics modules are miniature detectors that track energetic particles and photons. The numbers become dose and dose-rate. This helps the mission crew monitor radiation across orbit and during space weather events, assess dangers to onboard electronics and future human flights, and teach students detector physics, calibration, and space data processing.

Table 4. *Tertiary CubeSat Sensors*

Item	Description	Figures	Functions
1.	Assemble Geiger Counter Kit Counter Module Nuclear Radiation Detector		Makes the Geiger Tube Usable

2.4 Payload Development Timeline

The development of the CubeSat Payload involves planning and execution to ensure the success of its mission. The project is divided into three main phases: Planning and Design, Development, and Testing. The Planning and Design phase focuses on the system and design, payload requirements, and the mission goals. The Development phase focuses on testing and calibrating sensors, including assembling the parts, installing electronics, and programming. Lastly, the Testing phase and the integration with the communications and ground station to ensure the reliability of crucial components, such as the Cube Satellite's system functionality and the data the payload provides to the ground station.

Table 5. *Payload Project Timeline*

Phase	Description	Task	Time Frame
1	Planning and Design	Payload Mission Planning	Dec 17 – Dec 23 (Week 1)
		Define Mission Goals and concept	Dec 17 – Dec 23 (Week 1)
		Payload System Planning	Dec 27 – Jan 3 (Week 2)
		Finalize Payload Requirements, sensors, and modules (Purchasing)	Dec 27 – Jan 3 (Week 2)
		Payload Structural Design (PCB and 3D Design)	Jan 10 – Jan 16 (Week 4)
2	Development	Testing and Calibration of Sensors	Jan 3 – Jan 9 (Week 3)
		Assembly of Payload Sensors and Components	Jan 10 – Jan 16 (Week 4)
		Programming of CubeSat Mission	Jan 17 – Jan 23 (Week 5)
		Integration of parts	Jan 24 – Jan 30 (Week 6)
3	Testing	Payload 1st Testing	Jan 31 – Feb 6 (Week 7)
		Payload 2nd Testing	Feb 7 – Feb 13 (Week 8)
		Integration with Comms and Ground Station	Feb 14 – Feb 20 (Week 9)
		Payload Testing with Ground Station	Feb 21 – Feb 27 (Week 10)

Chapter 3 – Structure and Mechanism Subsystem

3.1 Structure and Mechanism Subsystem Timeline

This timeline outlines the planned activities, key milestones, and deliverables for the structural and mechanism design and fabrication phase of our CubeSat project. Our focus will be on ensuring the structural integrity, mechanism functionality, and adherence to standard CubeSat specifications and Standard Design Requirements.

Phase 1: Formulation and Outsourcing
Phase 2: Mechanism Design and Integration
Phase 3: Fabrication and Assembly
Phase 4: Testing and Verification

Table 5. *Structure and Mechanism Project Timeline*

Phase	Description	Task	Time Frame
1	Review 2U CubeSat Specification and Formulation of SDR document	Specifications and Characteristics of a 2U CubeSat and the Standard Design Requirement Document.	Jan 5 – Jan 12 (1 week duration)
	Procurement & Outsourcing of Materials	Bill of Materials (BOM) Draft, 3d Print Filament, and screws, etc.	Jan 5 – Jan 26 (4 weeks duration)
	Initial 3D CAD Structural Design	Preliminary CAD Model (File) for submission and creation.	Jan 7 – Jan 16 (1 week duration)
	Finite Element Analysis (FEA) - Static	FEA Report: Stress and Strain Analysis	Jan 19 – Jan 21 (1 week duration)
	Critical Design Review (Structural)	Structural CDR Presentation Calendar event	Jan 19 – Mar 20 (9 weeks duration)
2	Mechanism Concept Development	Mechanism Concept Sketches/Models	Jan 22 – Jan 26 (1 week duration)
	Detailed Mechanism CAD Design	Detailed Mechanism CAD Files	Jan 27 – Jan 30 (1 week duration)
	Dynamic/Kinematic Analysis	Mechanism Functionality Report	Feb 2 – Feb 5 (1 week duration)
	Mechanism Prototyping (Non-Flight Material)	Functional Prototype Video Demonstration	Feb 7 – Feb 16 (2 weeks duration)

3	Final Part Drawings Release	Manufacturing Drawings Package (File)	Feb 18 – Feb 24 (1 week duration)
	Structural Component Fabrication	All Structural Flight Parts	Feb 26 – Mar 9 (4 weeks duration)
	Mechanism Component Fabrication	All Mechanism Flight Parts	Feb 26 – Mar 9 (3 weeks duration)
	Initial Assembly and Fit Check	Assembled Structure Photo	Mar 11 (1 week duration)
	Mechanism Integration into Structure	Fully Assembled Structure with Mechanism	Mar 11 (1 week duration)
4	Structural Mass and Dimension Check	Mass & Dimension Compliance Report	Mar 10 (1 week duration)
	Vibration and Shock Testing	Vibration Test Report (Conducted at Place)	Mar 11 – Mar 12 (2 weeks duration)
	Vacuum (VAC) Testing	VAC Test Data Analysis	Mar 12 (2 weeks duration)
	Mechanism Deployment Functional Test	Final Deployment Test Video	Mar 12 – Mar 13 (1 week duration)
	Final Mechanical Verification Review	Final Approval for Bus Integration Calendar event	Mar 16 – Mar 18 (1 week duration)

Chapter 4 – Electrical Power System (EPS)

4.1 EPS Overview

The Electrical Power System (EPS) is responsible for generating, storing, regulating, and distributing electrical power to all subsystems of the 2U CubeSat. This component of the satellite ensures continuous and reliable operation by regulating the transfer of energy from solar panels, batteries, and power management systems. The EPS integrates hardware and simulation-based design to ensure the system operates seamlessly, reliably, and error-free during both sunny and eclipse conditions.

4.2 EPS Objectives

The objective of the EPS project is to design, develop, and evaluate a robust, efficient, and flight-ready power system that meets the specific requirements of the CubeSat mission. This involves selecting and assembling solar power generation systems, energy storage units, and power regulation circuits. Subsequently, the system must be tested, simulated, and developed on a PCB to ensure its ability to deliver consistent power across all mission scenarios and constraints.

4.3 EPS Team Organization and Responsibilities

Table 6 delineates the specific responsibilities assigned to each member of the Electrical Power System (EPS) team. It details the responsibilities of each individual in designing, analyzing, and documenting the CubeSat's power subsystem.

Table 6. *EPS Team Roles and Responsibilities*

Name	Role	Responsibilities
Archie Leo Henolos	EPS Team Lead	Oversee design, integration, review; approve documents and PCB
Chester Hahne Conde	Power Systems & PCB Engineer	Solar selection, power calculations, MPPT/DET/PPT, simulation support, PCB tasks
Kyle Confessor	Energy Storage, Regulation & PCB Engineer	Battery sizing/protection, DC-DC design, power flow analysis, PCB tasks
Kalex Rodas	Procurement & Documentation Specialist	Manage component procurement, track lead times, maintain records, prepare reports/manuals

4.4 EPS Development and Testing Timeline

Table 7 summarizes the sequential phases, key tasks, and timelines involved in the design, integration, and validation of the Electrical Power System (EPS) for the CubeSat.

Table 7. *EPS Project Timeline*

Phase	Description	Task	Time Frame
1	Mission & EPS Requirements	Define EPS needs. Review orbit & eclipse, estimate satellite power, define power modes & redundancy	Jan 5–7 (3 days duration)
	EPS Architecture	Decide power flow. Select EPS type (DET/PPT), main bus voltage, number of solar panels, battery chemistry	Jan 8–12 (1 week duration)
	Component Procurement	Secure EPS hardware. Purchase solar cells, batteries, controllers, DC-DCs; verify lead times	Jan 5–15 (1 week duration)
	Solar, Battery & Regulation Design	Design generation, storage & regulations. Solar power calc & selection, battery sizing & protection, charge controller & DC-DC design	Jan 5–20 (2 weeks duration)
2	Simulation & Verification	Validate EPS design. Power flow & worst-case analysis, battery DoD simulation, thermal estimation	Jan 21–25 (1 week duration)
	PCB Design & Review	Convert design to hardware. Finalize schematic, PCB layout, add test points, internal review, PCB fabrication (First week of Feb)	Jan 26–Feb 1 (1 week duration)
3	Assembly & Standalone Testing	Verify EPS standalone. Assemble PCB, bench test solar/battery, verify regulation & protections	Feb 2–8 (1 week duration)
	Integration Testing	Test EPS with CubeSat. Integrate with OBC/payload, measure consumption, startup/brownout tests, eclipse-mode simulation	Feb 9–15 (1 week duration)

Chapter 5 – Onboard Computer and Software

5.1 Onboard Computer (OBC) Overview

The onboard computer (OBC) in a CubeSat is the central control unit of the satellite, responsible for executing all core mission and housekeeping functions. It manages data acquisition from onboard sensors and payloads, processes and stores this data, controls satellite subsystems, and handles communication with the ground station. The OBC runs specialized, highly reliable onboard software designed to operate under the constraints of space—such as limited power, memory, and exposure to radiation—ensuring autonomous operation, fault detection and recovery, and successful mission execution throughout the CubeSat's lifetime.

5.2 OBC Team Organization

The following roles and timeline outline the planned development and implementation schedule for the CubeSat OBC, summarizing the key phases and milestones from design to testing and integration.

Table 8. *OBC Team Roles and Responsibilities*

Name	Role	Responsibilities
Reeven Niño E. Bugwat	Department Head / OBC Systems Lead	Responsible for the overall design, coordination, and technical direction of the onboard computer. This role defines system requirements, oversees both hardware and software development, coordinates with other CubeSat departments, manages the timeline, and ensures that the OBC integrates reliably with all subsystems.
Shawn Allen Dela Torre	OBC Software / Firmware Engineer	Develops the embedded firmware that runs on the onboard computer. This includes low-level initialization, communication drivers, data handling, fault management, and support for testing and debugging to ensure stable and reliable OBC operation.
Ailman Alexander Zamora	OBC Hardware Engineer	Designs and develops the custom onboard computer hardware, including the schematic and PCB layout. This role handles component selection, power and interface design, board fabrication preparation, and initial board bring-up and hardware debugging.

5.3 OBC Development and Integration Timeline

Table 9 outlines the project timeline for the On-Board Computer (OBC) subsystem, detailing each development phase, its corresponding tasks, and time frame from system planning to final integration and handover.

Table 9. *OBC Project Timeline*

Phase	Description	Task	Time Frame
1	System Planning & Requirements	Research bare-MCU architecture and define payload interfaces	Dec 27 – Jan 11 (2 weeks duration)
2	Schematic Design & MCU Selection	Finalize MCU, develop schematic, begin firmware concepts	Jan 12 – Jan 18 (1 week duration)
3	PCB Design & Design Freeze	Complete PCB layout, generate Gerber/BOM files	Jan 19 – Jan 25 (1 week duration)
4	PCB Fabrication & Firmware Bring-Up	Submit PCB, develop minimal firmware	Jan 26 – Feb 1 (1 week duration)
5	Firmware Development & Prep	Implement I/O drivers, prepare for assembly	Feb 2 – Feb 8 (1 week duration)
6	PCB Assembly & Power-On Testing	Assemble PCB, verify power/MCU boot	Feb 9 – Feb 15 (1 week duration)
7	Board Bring-Up & Interface Validation	Debug hardware/firmware, prepare payload integration	Feb 16 – Feb 22 (1 week duration)
8	Initial System Testing	Verify sensor communication and data flow	Feb 23 – Feb 29 (1 week duration)
9	Full System Integration	Run payload mission simulations, finalize firmware	Mar 1 – Mar 7 (1 week duration)
10	Final Review & Handover	Verify functionality, prepare for CubeSat integration	Mar 8 – Mar 14 (1 week duration)

Chapter 6 – Communication and Ground Station Subsystem

6.1 Communication Subsystem Overview

The Communication and Ground Station Subsystem is responsible for establishing reliable bidirectional data transfer between the CubeSat and the Earth-based control segment in support of weather monitoring, store-and-forward communication, and radiation dose monitoring payloads. The subsystem is designed in compliance with the CubeSat Design Specification (CDS) and relevant ECSS standards to ensure system compatibility, operational reliability, and regulatory compliance. Its primary function is to enable stable telemetry, telecommand, and payload data exchange throughout the mission.

The communication subsystem is designed to meet mission requirements, including a reliable uplink for command and configuration updates, sufficient downlink capacity for periodic payload data transmission, compatibility with amateur satellite frequency allocations, and low power consumption consistent with CubeSat electrical constraints. Based on expected payload data rates and operational scenarios, a low-to-moderate data rate communication architecture is selected to improve link reliability and reduce system complexity. This approach also simplifies ground station operations while ensuring consistent and efficient mission performance.

6.2 Communication and Ground Station Team Organization

Task distribution among team members is structured to ensure a balanced workload, clear technical ownership, and efficient progress across all mission phases. Assigned roles and responsibilities align with subsystem functions, supporting coordinated development and verification activities.

Table 10. *Comms and Ground Station Team Roles and Responsibilities*

Name	Role	Responsibilities
Joannah Marie S. Marinog	Communication & Ground Station Subsystem Lead	Oversees the end-to-end communication and ground station subsystem, ensuring requirements definition, system integration, testing, and documentation compliance.
Sheryn Paran Jarabe	RF Systems and Link Budget Engineer	Responsible for RF architecture design, link budget analysis, frequency selection, and verification of communication performance margins.
Josh Japitana	Antenna & Deployment Mechanism Engineer	Designs and validates the antenna system and deployment mechanism, including mechanical integration and electromagnetic performance.
Joshua Lemuel Dorias	Ground Station & Software Engineer	Develops and implements ground station hardware, tracking software, and communication protocols for reliable uplink and downlink operations.

6.3 Communication Subsystem Development Timeline

The development of the Communication and Ground Station Subsystem is organized into sequential phases of design, integration, and validation, conducted from January to early March.

Table 11. *Comms and Ground Project Timeline*

Phase	Description	Task	Time Frame
1	Subsystem Definition & Requirements	Finalize communication requirements and link budget assumptions	Jan 5–7 (1 week duration)
2	Radio Architecture Selection	Select transceiver, define uplink/downlink, modulation/protocol	Jan 8–14 (1 week duration)
	Link Budget Analysis	Analyze worst-case scenarios, complete link budget tables	Jan 8–14 (1 week duration)
3	Antenna System & Deployment	Select antenna type, design deployment mechanism	Jan 15–21 (1 week duration)
4	Procurement	Purchase antenna and components	Jan 22 (1 day duration)
	Development	Software development, antenna deployment mechanism	Jan 23–28 (1 week duration)
5	Detailed Design & Simulation	Antenna EM simulation, RF chain analysis, ground station definition	Jan 29–Feb 4 (1 week duration)
6	Ground Station Setup	Select hardware, configure tracking software, implement protocol	Feb 5–11 (1 week duration)
7	Integration with CubeSat Bus	Interface with OBC, power/command tests, payload packetization	Feb 12–18 (1 week duration)
8	Functional Testing	Antenna deployment, RF range testing, data integrity evaluation	Feb 19–25 (1 week duration)
9	System Validation	End-to-end testing, finalize parameters, draft documentation	Feb 26–Mar 3 (1 week duration)

Chapter 7 – Attitude Determination and Control System (ADCS)

7.1 ADCS Testing and Validation Timeline

Table 12 details the phased development timeline for the Attitude Determination and Control System (ADCS), covering design, procurement, fabrication, testing, and full CubeSat integration with specific tasks and time frames.

Table 12. *ADCS Project Timeline*

Phase	Description	Task	Time Frame
1	Design Parameters	Design spin wheel and magnetorquers parameters	Jan 5 – Jan 11 (1 week duration)
		Design/calculate mechanism dimensions	Jan 5 – Jan 11 (1 week duration)
		Select motors and materials	Jan 5 – Jan 11 (1 week duration)
		Finalize sensor selection	Jan 5 – Jan 11 (1 week duration)
2	Procurement	Procure required materials	Dec 23 – Jan 25 (4 weeks duration)
3	Fabrication	3D-print components, initial assembly	Jan 9 – Jan 28 (3 weeks duration)
4	Sensor Fusion	Gather code/libraries, develop test code	Jan 9 – Jan 16 (1 week duration)
5	Schematic Design	Research designs, create wiring	Jan 13 – Jan 22 (1 week duration)
6	PCB Design	Design/prototype PCB, initial assembly	Jan 16 – Jan 22 (1 week duration)
7	Algorithm & Mechanism Testing	Assemble electronics/mechanism, calibrate sensors	Jan 26 – Feb 15 (3 weeks duration)
8	Final PCB	Apply adjustments, manufacture PCB	Feb 1 – Feb 15 (2 weeks duration)
9	Mechanism Integration	Final adjustments, integrate mechanisms	Jan 13 – Feb 23 (6 weeks duration)
10	Full CubeSat Integration	Integrate all subsystems	Jan 15 – Jan 22 (1 week duration)

Chapter 8 – Master Project Timeline

8.1 Overall CubeSat Project Timeline

Table 13 presents the master project timeline for all CubeSat subsystems, detailing key phases, major deliverables, submission deadlines, team meetings, and testing objectives from initial planning through final documentation and integration.

Outputs / Progress Reports Expected | **Reports compiled and submitted by managers** | **Meeting updates / concerns** | **STRICTLY NO WORK**

Table 13. *Master Project Timeline*

Dates	Phase / Key Activities	Major Deliverables & Deadlines
Dec 15–19	Initial planning and proposal development	Draft proposal complete (all teams contribute mission overview).
Dec 19-20	**Timeline Proposal Deadline**	**Provide final documentation/report for compilation**
Dec 20-21	**Timeline Proposal submission**	Proposal compiled and submitted to instructor.
Dec 21–23	Review instructor comments; plan revisions	Review only, and maybe plan revisions especially on the components that needed replacement based on instructor's comment (to proceed w/ ordering)
Dec 24-Jan 2	***Christmas & New Year***	
Dec 27–Jan 3	Ordering only	**Procurement list locked; place orders** (payload sensors, EPS solar/batteries, ADCS materials, structure outsourcing).
Jan 3–5	Execute revisions/adjustments	Create adjustments from instructor's comments
Jan 5–10	System requirements & paperwork; continue orders	**System requirements docs** (comms/EPS/ADCS/structure SDRs); programming outlines; procurement tracking.
Jan 12	**Team meeting: updates**	Progress review; concerns
Jan 13–17	Continue systems; adjustments from Jan 12 meeting	Initial designs iterated (payload struct, comms antenna concepts, EPS architecture, ADCS schematics).
Jan 19	**Sinulog - no meeting**	Independent progress
Jan 20	**Team meeting**	Progress review; comments addressed; other
Jan 20–23	Continue progress; adjustments from Jan 20 mtg	**Jan 24 prep**: System designs/charts/diagrams ready.
Jan 24	**Presentation of progress outputs**	**Major deadline: Present designs, CAD, diagrams (all subsystems).**

		Teams must send presentation and documentation for compilation
Jan 25	**Submission of Progress Report**	**Compilations done. Submit to instructor**
Jan 26–Feb 3	Progress continuation & initial fabrication	PCBs/schematics finalized (EPS/ADCS/comms); mechanism prototypes start; payload assembly.
Feb 4	**Team meeting: updates**	Fabrication status review.
Feb 5–10	Fabrication & component integration	Standalone tests (EPS power, payload sensors, comms protocols); 3D prints/assemblies.
Feb 11	**Team meeting: updates**	Integration plans confirmation.
Feb 12–14	Adjustments & refinements	Make adjustments from the meeting comments; continue progress
Feb 15	**Team meeting: fabricated PCB outputs**	**Show PCBs & prototypes (EPS/ADCS/comms); test reports. Must submit documentations or report for compilation**
Feb 16	**Submission of Documentation / Progress**	**Compilations done. Submit to instructor**
Feb 16–23	Subsystem integration testing	EPS/CubeSat integration; comms bus tests; ADCS mechanism calibration; structure prototyping.
Feb 24	**Testing Result Submission & Presentation**	**Present with team heads. Must provide documentation and reports for compilation**
Feb 24-25	**Submit Testing Result to Instructor**	**Compilations done. Submit to instructor**
Feb 24 – Mar 2	Environmental testing & validation	Payload/comm GS tests; RF range/deployment; EPS final docs; comms end-to-end; structure fabrication starts. **Mar 3: Draft paper sections**.
Mar 3	**Testing Result Submission & Presentation**	**Present with team heads. Must provide documentation and reports for compilation**
Mar 3-4	**Submit Testing Result to Instructor**	**Compilations done. Submit to instructor**
Mar 4–13	Full **STRUCTURE** assembly & qualification tests	Structure fab/assembly; checks; mechanism deployment.
March 14	**Team meeting: progress**	**Show progress reports, outputs, etc.**
Mar 14–20	Final reviews & bus integration approval	**Mar 18: Mech verification review**. **Mar 20: Structural CDR & full CubeSat readiness** (10am initial manual deadline).
Mar 16–19	**Each team manual making**	Subsystem manuals drafted (EPS, comms, payload, ADCS, structure interfaces).

Mar 20	**Manual Initial Draft Submission**	**To be checked by department heads**
Mar 20	**Meeting for manual adjustments**	Review initial manuals; assign revisions.
Mar 21	Manual revisions	Teams implement meeting feedback. Must provide initial manual (each team)
Mar 23	**Initial manual submission**	**Compiled manuals submitted to instructor for review.**
Mar 24–26	Adjustments based on sir's comments	Address instructor feedback across all manuals.
Mar 27	**Meeting presentation of manual adjustments**	**Final comments/concerns review. Must provide revised manual for compilation**
Mar 28	**Submission of final manual**	**Compilations done. Updated manual submitted to instructor**
Mar 29–30	Revisions if necessary	Polish any remaining issues.
Mar 30	**Final manual submission**	**Each team must submit their final manual**
Mar 30-31	**Compiled final manual submission**	**Project documentation compiled**

Note: Each team timelines may have earlier internal deadlines (they can complete ahead of schedule but never later). This ensures that they can present timely progress updates and reports when required.

This document is an edited and compiled version of the individual subsystem documents submitted by each team head.

Edited and compiled by:

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